Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 11: Collections

Sets, Subsets, Power Sets, and Cartesian Products

Chartrand & Zhang, Sections 2.1 and 2.3

Name: _____ Date: _____

Where we are

Everything from here — sets, relations, functions, graphs — is built on the logic you now have. Next week's set operations are *defined* using $\wedge$, $\vee$, and $\sim$, so the laws you proved in Week 1 will reappear as laws about sets.

By the end of today you should be able to

1. Describe a set by listing or by a defining property.
2. Tell $\in$ from $\subseteq$ and use each correctly.
3. Write the power set of a small set and give its size.
4. Form a Cartesian product and give its size.

1. Describing a set

A **set** is a collection of objects, called its **elements**.

By listing	$A = \{1, 3, 5, 7, 9\}$
By property	$A = \{x \in \mathbb{Z} : x \text{ is odd and } 1 \leq x \leq 9\}$

$\emptyset$	the **empty set** — no elements
$\lvert A \rvert$	the **cardinality** of A — how many elements
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$	naturals, integers, rationals, reals
$a \in A$	a is an **element** of A
$A \subseteq B$	A is a **subset** of B

A subset claim is a quantified implication

$A \subseteq B$ says exactly

$$\forall x,\ \big(x \in A \Rightarrow x \in B\big).$$

To *prove* it, take an arbitrary $x \in A$ and show $x \in B$. To *disprove* it, exhibit one element of A not in B. That is Wednesday's asymmetry, applied on day two.
It also settles something that looks strange: $\emptyset \subseteq B$ for *every* set B, because the hypothesis $x \in \emptyset$ is never satisfied. Vacuously true — Lecture 4.

2. $\in$ versus $\subseteq$

$\in$ relates an **element** to a set. $\subseteq$ relates a **set** to a set.

Let $D = \{1, 2, \{2\}\}$. Then D has 3 elements: 1, 2, and the set $\{2\}$.

Claim	T/F	Why
$2 \in D$	T	2 is listed as an element.
$\{2\} \in D$	T	The set $\{2\}$ is also listed.
$\{2\} \subseteq D$	T	Its only element, 2, is in D.
$\{\{2\}\} \subseteq D$	T	Its only element, $\{2\}$, is in D.
$3 \in D$	F	Not listed.

Braces carry meaning

$\emptyset$ is the empty set. $\{\emptyset\}$ is a set containing one thing — the empty set — so $|\{\emptyset\}| = 1$, not 0.
Whenever you see nested braces, count the elements at the **outermost** level first.

3. Power sets

Definition 2.15

The **power set** of A, written $\mathcal{P}(A)$, is the set of *all* subsets of A:

$$\mathcal{P}(A) = \{B : B \subseteq A\}.$$

Since $\emptyset \subseteq A$ and $A \subseteq A$ always, both are elements of $\mathcal{P}(A)$.

- $\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
- $\mathcal{P}(\emptyset) = \{\emptyset\}$ so $|\mathcal{P}(\emptyset)| = 1$, not 0.

Theorem 2.17

If $|A| = n$ for a nonnegative integer n, then $|\mathcal{P}(A)| = 2^n$.

$\lvert A\rvert$	0	1	2	3	4	5
$\lvert\mathcal{P}(A)\rvert$	1	2	4	8	16	32

You counted this on Day 1

Why 2^n? Building a subset means deciding, for each of the n elements, in or out — two independent choices, n times.
That is the four-character binary password from Lecture 1 with a different story attached. Chartrand proves it formally as Theorem 4.12, by induction, later in the term.

4. Cartesian products

Definition 2.37

For sets A and B, the **Cartesian product** is

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\},$$

the set of all *ordered pairs* with first coordinate in A and second in B.

Ordered matters: $\{a, b\} = \{b, a\}$, but $(a, b) \neq (b, a)$ when $a \neq b$.

For $A = \{1, 2\}$ and $B = \{x, y, z\}$:

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$$

$|A| = 2$, $|B| = 3$, and $|A \times B| = 6$.

Counting rule

For finite sets, $|A \times B| = |A| \cdot |B|$.

Is $A \times B = B \times A$? No — the pairs are ordered. Their *sizes* are equal, though.

Where you have already seen this

$\mathbb{R} \times \mathbb{R}$ is the coordinate plane. A database table with columns drawn from A and B has rows in $A \times B$. And every counting argument from Lecture 1 was a Cartesian product in disguise: four binary characters is $\{0, 1\} \times \{0, 1\} \times \{0, 1\} \times \{0, 1\}$, of size 2^4.

Exit ticket

1. Let $A = \{1, \{1\}\}$. Is $\{1\} \in A$? Is $\{1\} \subseteq A$? Explain each briefly.

2. If $|A| = 4$ and $|B| = 3$, give $|\mathcal{P}(A)|$ and $|A \times B|$.

Before Monday

Read Section 2.2, "Set Operations and Their Properties," pp. 60–70. You will recognise the laws — they are Week 1's, rewritten for sets.

Self-check (not collected). Section 2.1, Exercises 1, 3, 5, and 11 — odd-numbered, answers in the back.

Coming up

Homework 4 is due tonight, Friday, September 11, 11:59 p.m.
Quiz 4 is Monday, September 14 — 15 minutes on Sections 3.1 and 2.1.
Homework 5 covers Sections 3.1, 2.1, and 2.3, due Friday, September 18.

Questions I still have